S&W framework refresh: kij refit on the S&W 1992 equilibrium procedure

Coefficients, fit quality and comparison with the published set

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1 Summary

The correlations published in Burgoyne and Nielsen (2026), *Fluid Phase Equilibria* 114824, were regressed with a binary gas-water solve that treats the vapour as pure, dry gas at total pressure. Søreide and Whitson (1992) fitted their correlations with one equilibrium in which the aqueous phase uses k_{ij}^{AQ} and the non-aqueous phase uses k_{ij}^{NA}, with the vapour carrying its equilibrium water (their Eqs. 10a/10b). This note refits the aqueous-phase BIPs, re-checks the non-aqueous BIPs and refits the embedded salinity correction on that procedure, using the same data, exclusions, temperature window, functional forms and fitting routines as the paper. Only the equilibrium procedure changes. The published coefficients are unchanged and remain correct only inside the dry-vapour solve they were fitted with.

Findings, freshwater solubility on the paper's point set (Table 1):

- In a procedure-1 calculation the published k_{ij}^{AQ} underpredict solubility wherever the vapour holds appreciable water: bias −12 to −29% at 150-200 °C for CO_2, H_2S, CH_4 and H_2. The refit removes it (−6 to +3%).
- The refit shifts every pointwise k_{ij}^{AQ} downward (1,561 of 1,561 points), by less than 0.001 at or below 60 °C and by 0.013-0.086 (gas mean) at 150-200 °C, several times the pointwise scatter (MAE 0.006-0.035). Measurement noise cancels in this paired difference.
- Scored on procedure 1, the S&W 1992 correlations are closer to the refit than the paper reported: CO_2 15.3% vs 12.4%, H_2S 8.0% vs 6.8% (excluding 18 suspect points, Section 6), CH_4 3.9% vs 3.9%. The paper's S&W column (23.0, 51.3, 5.8%) scored S&W's coefficients in the dry-vapour solve, which penalised them.
- k_{ij}^{NA}: no change justified for any gas under the paper's 1.1 pp rule. All published constants retained.
- Helium (a post-publication extension, not in the paper): the procedure-1 refit moves k_{ij}^{AQ} by −0.0002 on average, because the helium correlation was already fitted on a gas-fugacity basis; freshwater MARE 6.3% ($n = 130$), brine 12.8% ($n = 48$). Section 5.
- Brine: the refitted salt correction differs from the published one by 0.4-2.9% in mean salt factor; brine MAREs are within the scatter of the published set once the freshwater bias is accounted for (Section 4.4).

2 Procedure and method

Three ways of computing gas solubility with the two S&W BIP sets appear in practice. They differ only in how much water the non-aqueous phase carries when the aqueous gas fugacity is equated to the vapour's:

Procedure	Non-aqueous phase	Used by
0, dry vapour	pure gas at total pressure	published paper (regression and MARE)
1, S&W Eqs. 10a/10b	one equilibrium; aqueous phase k^{AQ}, non-aqueous phase k^{NA}	S&W 1992; this refit
2, dual flash	k^{AQ} in every phase for the solubility flash	Whitson & Brulé (2000) shortcut for single-BIP EOS software

Whitson and Brulé (2000, §9.3) give procedure 2 for cases where only gas solubility or only water content is needed and note that predicting both requires an EOS implementation with phase-specific BIPs, which is procedure 1.

EOS. PR79, S&W water alpha with NaCl molality (S&W Eq. 9), applied to water in both phases (one pure-component set; only the BIPs are phase-specific). Component properties: S&W Table 3 (H_2 and He from NIST), as in the paper.

Solver. Successive substitution on $K_i = \phi_i^L(x; k^{\mathrm{AQ}})/\phi_i^V(y; k^{\mathrm{NA}})$ with the closed-form binary Rachford-Rice, falling back to a nested bracketed solve. Verification against an independent implementation (Younus 2026, Table 4, seven binaries with S&W 1992 BIPs): water contents for both procedures 1 and 2 agree to all printed digits, and the procedure-2/procedure-1 solubility ratios agree to ±0.01 percentage points; fugacity residuals are below 2×10^{-13} in $\ln f$.

Regression. Pointwise k_{ij}^{AQ} per freshwater point with k_{ij}^{NA} at its recommended value; correlations fitted by L1 (the paper's routine and forms); k_{ij}^{NA} re-checked with the paper's grid search and rows; embedded Δk_{ij} fitted with the paper's routine to the same Sechenov targets (Dubessy et al. 2005 −0.011 for CO_2, Akinfiev 2016 +0.019 for H_2S, S&W Eq. 8 otherwise) on the same $T/P/m$ grids. Because k_{ij}^{NA} did not change, no further $k^{\mathrm{AQ}}/k^{\mathrm{NA}}$ iteration was needed.

3 Coefficients

3.1 Aqueous-phase BIP, freshwater

Forms as published; T in K, $T_r = T/T_c$.

$$\begin{aligned} \text{cubic:} \quad & k = c_0 + c_1 T + c_2 T^2 + c_3 T^3 \\ \text{exponential:} \quad & k = a/T + b\,\exp(c/T) + d \\ \text{linear:} \quad & k = a + bT \\ \text{rational:} \quad & k = (A + T_r)/(B + C\,T_r) \end{aligned}$$

Gas	Form	Coefficients	n	MAE k	max $\lvert r\rvert$	T fit (°C)
CO_2	cubic	c_0=−1.550211, c_1=9.32272×10^{-3}, c_2=−1.977702×10^{-5}, c_3=1.527732×10^{-8}	606	0.0087	0.164	0-200
H_2S	exp	a=−82.78098, b=1.739491×10^{6}, c=−8585.063, d=0.2518306	405	0.0070	0.204	21-200
CH_4	rational	A=−2.197678, B=0.7150154, C=0.894503; T_c=190.60	115	0.0076	0.388	25-171
N_2	linear	a=−1.621096, b=3.295273×10^{-3}	127	0.0104	0.034	25-200
H_2	rational	A=−16.01983, B=0.7312363, C=0.681362; T_c=33.145	154	0.0350	0.263	0-179
C_2H_6	rational	A=−1.271348, B=−0.02927137, C=1.637183; T_c=305.40	95	0.0063	0.057	2-200
C_3H_8	rational	A=−1.156778, B=0.208946, C=1.810663; T_c=369.80	59	0.0015	0.024	6-149
He	rational	A=−71.6155, B=−0.164407, C=0.5017371; T_c=5.1953	130	0.0184	0.124	20-163

The H_2S exponential and the He and CH_4 rationals have strongly correlated parameters: individual coefficients moved a long way from the published values (H_2S b from 1.349×10^3 to 1.739×10^6) while the curves moved by the amounts in Section 4.2. Use all digits given. Heavier hydrocarbons (iC_4 and above) keep S&W Eqs. 11-12, which S&W fitted on procedure 1.

Check values, freshwater k_{ij}^{AQ} at 25 / 80 / 150 / 200 °C: CO_2 −0.12379 / −0.05152 / +0.01103 / +0.05158; H_2S −0.02582 / +0.01747 / +0.05889 / +0.09980; CH_4 −0.29959 / −0.14536 / +0.00830 / +0.09700; N_2 −0.63861 / −0.45737 / −0.22670 / −0.06194; H_2 −1.02393 / −0.67140 / −0.34499 / −0.16683; C_2H_6 −0.18807 / −0.06170 / +0.05101 / +0.11085; C_3H_8 −0.21005 / −0.10412 / −0.00549 / +0.04858; He −0.49694 / −0.10726 / +0.24159 / +0.42735. Equilibrium check values (procedure 1, x_{gas} and y_{H_2O} at three states per gas) are in the accompanying `check_values.csv`.

3.2 Non-aqueous-phase BIP

Unchanged from the paper: CO_2 0.1896, H_2S 0.161, CH_4 0.4850, N_2 0.4778, H_2 0.468, C_2H_6 0.4920, C_3H_8 0.5525, nC_4 0.5091, C_5+ 0.500; He 0.468 (H_2 surrogate, no He water-content data). Table 2 gives the procedure-1 evidence.

3.3 Embedded salinity correction

$k_{ij}^{\mathrm{AQ}}(T,m) = k_{ij}^{\mathrm{AQ}}(T) + (a_0 + a_1 T_r + a_2 T_r^2)\, m$ [$+ (b_0 + b_1 T_r)\, m^2$ for CO_2], m = NaCl molality. The S&W brine water alpha is applied with it. MAE and max are against the Sechenov targets on the fitting grid.

Gas	T_c (K)	a_0	a_1	a_2	b_0	b_1	MAE x (%)	max (%)
CO_2	304.20	0.0408481	−0.0789861	0.0502757	0.00676611	−0.00735569	2.92	15.8
H_2S	373.20	0.0474380	−0.0916139	0.0498240			2.35	19.1
CH_4	190.60	0.134391	−0.134258	0.0408428			0.98	3.7
N_2	126.10	0.219484	−0.148600	0.0306167			1.04	4.2
H_2	33.145	0.376712	−0.0647304	0.00315541			0.66	2.4
C_2H_6	305.40	0.0777696	−0.123360	0.0625866			1.09	3.6
C_3H_8	369.80	0.0580567	−0.111055	0.0692266			1.40	3.3
nC_4H_{10}	425.20	0.0456416	−0.100089	0.0729267			1.75	3.4
He	5.1953	0.363020	−0.00963936	7.09302×10^{-5}			0.56	1.3

C_3H_8 and nC_4H_{10} are now fitted on their own freshwater BIPs (the paper carried them over from the MC-3 fit). The CO_2 and H_2S maxima sit at high temperature, at the 50 bar and 500 bar ends of the grid: on procedure 1 a pressure-independent Δk_{ij} cannot follow the change in vapour water content with pressure as closely as it could in the dry-vapour solve (the published fit gave 1.56% and 1.99% MAE).

4 Fit quality

4.1 Freshwater solubility

Columns: **Pub-P0** published coefficients in the published dry-vapour solve (reproduces the paper exactly); **Pub-P1** published coefficients in a procedure-1 calculation; **SW-P1** S&W 1992 aqueous and non-aqueous BIPs (Eq. 17 for H_2S) on procedure 1, as they were fitted; **New-P1** this refit. Evaluation set is the paper's (exclusions, 0-200 °C; H_2 excludes Gillespie 1980 as in the paper). Five CO_2 points cannot be evaluated on procedure 1 (Section 6).

Table 1: Freshwater x_{gas} MARE (%), with bias (%) in brackets.

Gas	n	Pub-P0	Pub-P1	SW-P1	New-P1
CO_2	611 (606)	16.4 (+10.7)	14.9 (−1.0)	15.3 (−6.3)	**12.4** (+4.8)
H_2S	405	38.5 (+30.4)	38.2 (+13.9)	30.8 (+20.6)	31.9 (+25.7)
H_2S, excl. 18 suspect	387	13.8	13.8	8.0	**6.8**
CH_4	115	4.8 (−0.3)	4.7 (−3.0)	3.9 (−0.6)	**3.9** (−0.7)
N_2	127	5.7 (+1.6)	5.6 (−0.4)	7.2 (+3.7)	**5.3** (+1.2)
H_2	142	**10.2** (+3.0)	12.0 (−1.1)	-	11.3 (+2.3)
C_2H_6	95	12.9 (+4.6)	9.2 (−2.4)	11.3 (−5.5)	**8.4** (+0.8)
C_3H_8	59	5.0 (+1.4)	4.8 (−1.2)	8.9 (+4.8)	**3.4** (+1.2)
He	130	(9.7)[a]	6.3 (+0.1)	-	6.3 (+0.1)

[a] The He correlation was fitted and published on a gas-fugacity basis (helium partial pressure), which is already close to procedure 1; its dry-vapour score is shown for completeness only.

The temperature trend is where the procedure matters. MARE (bias) at 150-200 °C:

Gas	n	Pub-P0	Pub-P1	SW-P1	New-P1
CO_2	113	30.0 (+23.3)	20.7 (−20.4)	8.4 (−6.0)	9.0 (−2.4)
H_2S	103	31.9 (+20.7)	28.8 (−28.8)	8.1 (+2.2)	6.8 (+2.9)
CH_4	21	12.7 (−2.9)	12.8 (−12.2)	9.3 (−5.3)	9.2 (−5.9)
N_2	11	6.9 (+3.5)	5.8 (−5.8)	5.3 (−5.3)	3.3 (−0.8)
H_2	20	9.1 (+3.4)	17.2 (−13.3)	-	11.6 (−2.9)
C_2H_6	26	14.1 (+8.5)	6.9 (−5.2)	6.0 (−3.2)	5.8 (+0.9)

At or below 60 °C the published and refitted columns agree within 2 pp for every gas (H_2S aside, where the suspect points of Section 6 dominate): the vapour is nearly dry and the procedure barely matters. H_2 is the one gas where the refit scores worse overall than the published solve (11.3% vs 10.2%); the difference is at 100-200 °C, where the H_2 data are sparse (47 points) and the pointwise scatter is the largest of any gas (MAE 0.035).

4.2 Size of the shift

Pointwise k_{ij}^{AQ} (procedure 1 minus published, same measured x), gas mean by temperature band:

Gas	≤60 °C	60-100	100-150	150-200
CO_2	−0.0008	−0.0040	−0.0114	−0.0528
H_2S	−0.0004	−0.0021	−0.0109	−0.0483
CH_4	−0.0001	−0.0006	−0.0043	−0.0185
N_2	−0.0003	−0.0042	−0.0061	−0.0266
H_2	−0.0003	−0.0083	−0.0330	−0.0863
C_2H_6	−0.0007	−0.0006	−0.0061	−0.0126
C_3H_8	−0.0001	−0.0008	−0.0040	-

4.3 Water content

Table 2: y_{H_2O} MARE (%) on procedure 1 with the refitted k_{ij}^{AQ}; rows as the paper's k_{ij}^{NA} analysis.

Gas	n	k^{NA}	MARE	S&W value	optimum k^{NA} (MARE)	paper (dry solve, MC-3 alpha)
CO_2	208	0.1896	31.7	31.7	0.190 (31.7)	35.1
H_2S	162	0.161	6.7	7.1 (Eq. 17)	0.147 (6.4)	8.6
CH_4	93	0.4850	7.1	7.1	0.495 (6.7)	10.2
N_2	31	0.4778	8.1	8.1	0.537 (7.5)	7.2
H_2	14	0.468	4.2	-	0.499 (4.1)	4.5
C_2H_6	58	0.4920	19.0	19.0	0.476 (18.1)	21.1
C_3H_8	11	0.5525	5.5	5.5	0.557 (5.4)	4.1

The largest re-optimisation gain is 0.9 pp (C_2H_6), below the paper's 1.1 pp threshold, so every constant stays. On procedure 1 the paper's H_2S change (0.161 replacing S&W Eq. 17) gains 0.44 pp against 3.1 pp in the dry solve; it is retained for continuity, but by the paper's own rule it would no longer be justified.

4.4 Brine

Gas	n	Pub-P0	Pub-P1	SW-P1	New-P1
CO_2	109	17.3 (+10.0)	18.0 (+7.3)	18.4 (−15.2)	17.1 (+11.6)
H_2S	100	15.4 (+8.1)	7.6 (−2.7)	64.9 (−64.9)	11.0 (+4.3)
CH_4	32	5.6 (+2.4)	5.1 (+1.7)	11.9 (−9.8)	6.5 (+3.3)
N_2	45	4.6 (−2.8)	4.8 (−3.3)	4.3 (−1.3)	4.2 (−1.5)
H_2	47	4.1 (−0.3)	3.9 (−0.8)	-	5.8 (−2.5)
He	48	13.0 (+3.4)	12.8 (+3.2)	-	12.8 (+3.2)

The published coefficients score better on H_2S, CH_4 and H_2 brine in a procedure-1 calculation than the refit does. Splitting each prediction into its freshwater baseline and its salt factor shows why: the mean salt factors of the two sets agree within 0.4-2.9% per gas, and the difference comes from the freshwater baseline. For Suleimenov (1994) H_2S at 155 °C the published baseline is 28% low on procedure 1, which offsets a salt correction that is too weak at that temperature (brine error −9.7%); with the corrected baseline the refitted salt correction gives +17.1%. The refit exposes a high-temperature weakness in the H_2S Sechenov model rather than introducing an error.

Sensitivity: defining the Sechenov target as an activity coefficient at equal gas fugacity, instead of the paper's ratio at equal T and total P (which is also how the measured k_s were extracted), changes every brine MARE above by at most 0.3 pp. The total-pressure definition is used.

5 Helium

Helium was added to the framework after publication (October 2026) and is not part of the peer-reviewed paper. It is documented here in full because the procedure-1 basis is the first one on which its coefficients are recommended alongside the other gases.

Properties. NIST (Ortiz-Vega et al. 2019 equation of state): $T_c = 5.1953$ K, $P_c = 0.228323$ MPa, $\omega = -0.38354$, $T_b = 4.2238$ K.

Data. 130 freshwater points from five sources, 20-163 °C and 1-1,000 bar, in `shared/data/he_solubility.csv` with DOIs (fit and validate flags as in the October fit): Abrosimov and Lebedeva (2013, evaluated values; 105 points, 50-80 °C, 1-1,000 bar), Gardiner and Smith (1972; 12, 20-50 °C), Gerth (1983; 8, 25 °C), Potter and Clynne (1978; 2, 104-149 °C) and Pray et al. (1952; 3, 163 °C). The October fit targeted pointwise k_{ij} at the PR fugacity of pure helium at the helium partial pressure, which already approximates procedure 1; the vapour water fraction on procedure 1 ranges from 1.5×10^{-4} to 0.50 across these points.

Aqueous-phase BIP. Rational form, coefficients in Section 3.1 ($A = -71.6155$, $B = -0.164407$, $C = 0.5017371$). The individual coefficients differ from the October values (-71.6152, -0.394962, 0.505807), but the curve moves by -0.0002 on average (-0.0002 to $+0.0008$) over the data: the rational parameters are strongly correlated.

Source	n	T (°C)	P (bar)	Pub-P0	New-P1 (bias)	max
Abrosimov & Lebedeva 2013	105	50-80	1-1,001	9.5	6.8 (+0.2)	12.6
Gardiner & Smith 1972	12	20-50	101-608	2.4	2.4 (+0.7)	6.2
Gerth 1983	8	25	51-405	2.8	2.8 (−0.3)	5.3
Potter & Clynne 1978	2	104-149	11-13	39.0	2.7 (+2.7)	2.9
Pray et al. 1952	3	163	14-27	46.2	15.9 (−6.2)	27.0

MARE in %. The large Pub-P0 errors at low pressure and high temperature are the dry-vapour solve, not the correlation: at 104-163 °C and 11-27 bar the vapour is 10-50% water. On Abrosimov and Lebedeva the residuals carry a pressure trend (overprediction at 1-10 MPa, underprediction at 70-100 MPa), the structured bias of a $k_{ij}(T)$-only form. Not validated above 163 °C.

Non-aqueous-phase BIP. 0.468, the H_2 value used as a surrogate. No helium water-content data were found, so water content of helium-rich gas is unvalidated.

Embedded salinity correction. Coefficients in Section 3.3, fitted to S&W Eq. 8 Sechenov targets ($T_b = 4.2238$ K) on the H_2 grid restricted to 25-100 °C. Eq. 8 matches measured helium k_s in NaCl to 0.0067 kg/mol mean absolute deviation ($n = 48$). Brine MARE is 12.8% ($n = 48$, bias +3.2%), dominated by Gardiner and Smith's 100 °C polynomial points (+20 to +43%), which Potter and Clynne show are wrong at that temperature.

6 Data notes

- CO_2, Wang (2021), 5 points at 160-200 °C and 5-10 bar: the stated pressure is at or below the water vapour pressure (6.2 bar at 160 °C, 15.5 bar at 200 °C), so it must be CO_2 partial pressure. No aqueous phase exists on procedure 1 at that total pressure; excluded from the fit and the procedure-1 MAREs.
- H_2S, Selleck (1952) at 37.8 °C, 6 points: tabulated x_{H_2S} falls from 0.0037 to 0.0011 as pressure rises from 7 to 27 bar, which no solubility isotherm does; every model is off by about 1,200%. Suleimenov (1994) at or below 60 °C, 12 points, about 250% for every model. These 18 points are in the paper's set and set its H_2S MARE (38.5%; 13.8% without them). Flagged for source check, not removed here.
- Nine H_2S points near the water-saturated mixture critical locus (Selleck 1952, Koschel 2013; 104-138 °C, 155-300 bar) need an edge-refined root search on procedure 1; all converge.

7 Limitations

- Binary data only, as in the paper. Procedure 1 makes the coefficients transferable to multicomponent and oil systems in principle; that is not tested against data here.

- The forms and the temperature limits are the paper's; the correlations should not be extrapolated beyond the fitted ranges listed above (200 °C at most).
- Brine targets rely on the paper's Sechenov models; the H_2S model is weak at 155 °C (Section 4.4).

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